

# Incident Response Tools and Automation

Estimated time needed: 7 minutes

## Learning objectives

After completing this reading, you will be able to:

- Discuss real-time tools for incident detection and alerting
- Explain the functions of platforms for gathering and sharing threat intelligence
- Describe how platforms manage incident response workflows and documentation
- Explain how SOAR tools automate and orchestrate security tasks
- Discuss the role of EDR solutions in monitoring and responding to endpoint threats
- Illustrate the importance of analyzing network traffic for detecting threats
- List the tools that assist with regulatory compliance and reporting

## Incident detection and alerting

Incident detection and alerting tools help to spot and alert security incidents in real time. Some of the most used are:

- **Splunk:** Offers real-time monitoring and alerting capabilities
- **Microsoft Sentinel:** Cloud-native solution integrating intelligent security analytics with threat intelligence
- **IBM QRadar:** Integrates with various data sources to offer comprehensive threat detection and alerting
- **Snort:** Open-source intrusion detection system monitors network traffic for suspicious activity

## Threat intelligence platforms

Threat intelligence platforms are crucial for gathering, analyzing, and sharing threat information. They include:

- **ThreatConnect:** Combines threat intelligence, analytics, and orchestration
- **IBM X-Force:** Exchange offers crucial threat intelligence data and insights to help organizations avoid threats
- **AlienVault Open Threat Exchange (OTX):** Open-source platform for sharing threat intelligence
- **Anomali ThreatStream:** Collects and organizes threat data from various sources to provide actionable intelligence

## Incident response platforms

Incident response platforms help to manage workflows and documentation during incidents. They include:

- **CrowdStrike Falcon:** Provides endpoint detection and response capabilities and supports incident response workflows
- **IBM Resilient:** Helps manage and automate incident response processes
- **Cortex XSOAR:** SOAR platform that integrates incident response workflows
- **TheHive:** Open-source platform that offers case management and collaboration features

## Security orchestration, automation, and response (SOAR)

SOAR tools help to automate tasks and integrate multiple security solutions. They include:

- **Splunk Phantom:** Integrates with various security tools to automate and orchestrate incident response
- **IBM Resilient:** Provides automation and orchestration to streamline incident response
- **DFLabs IncMan:** Offers automation, orchestration, and response capabilities
- **Cortex XSOAR:** Combines automation with orchestration to improve incident response efficiency

## Endpoint detection and response (EDR)

EDR tools help to monitor end-user devices for suspicious activity and incidents. They include:

- **Microsoft Defender:** Offers advanced threat protection and EDR capabilities
- **CrowdStrike Falcon:** Provides comprehensive EDR features for detecting and responding to endpoint threats
- **SentinelOne:** Delivers real-time endpoint protection and response
- **OSSEC:** Open-source host-based intrusion detection system with EDR capabilities

## Network traffic analysis

Network traffic analysis tools help to monitor and analyze network traffic for anomalies and potential threats. They include:

- **SolarWinds NetFlow Traffic Analyzer:** Monitors and analyzes network traffic to detect anomalies
- **ManageEngine NetFlow Analyzer:** Offers real-time network traffic analysis and monitoring
- **Wireshark:** Open-source network protocol analyzer that captures and analyzes network traffic
- **Plixer Scrutinizer:** Helps identify and mitigate network threats through traffic analysis

## Compliance and reporting tools

Compliance and reporting software solutions assist with regulatory compliance and reporting on security status. They include:

- **AuditBoard:** Compliance management platform that helps organizations manage audits and compliance

- **Netwrix Auditor:** Provides visibility into IT infrastructure changes and supports compliance reporting
- **Vanta:** Automates security monitoring and compliance reporting for various standards
- **OpenSCAP:** Open-source tool for automated compliance checking and reporting

## Summary

In this reading, you learned that:

- Incident detection and alerting tools helps with real-time monitoring and alerting of security incidents.
- Threat intelligence platforms gather, analyze, and share threat information.
- Incident response platforms are systems for managing and automating incident response workflows.
- Security orchestration, automation, and response tools help to automate and orchestrate security tasks and responses.
- Endpoint detection and response help monitor and respond to threats on end-user devices.
- Network traffic analysis is a tool for monitoring and analyzing network traffic for anomalies.
- Compliance and reporting tools are software for managing regulatory compliance and security reporting.

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